

SOFTWARE REQUIREMENTS SPECIFICATION

Feature Flag & Dynamic Config Manager *Problem Statement #43*

1. Introduction

1.1 Purpose

This document specifies the proposed requirements for a Feature Flag & Dynamic Config Manager. It describes the system's intended capabilities, users, interfaces, constraints, and quality requirements so that the project can be designed and verified consistently.

1.2 Scope

The system is a centralized platform for managing feature flags and dynamic configuration. It is intended to let teams define flags, target selected users or cohorts, control percentage-based rollouts, configure environment-specific values, and propagate changes to connected client SDKs without requiring application redeployment.

1.3 Definitions

Feature flag: A named configuration that determines whether an application feature is enabled or disabled.

Targeting rule: A condition used to decide which users or cohorts receive a flag state.

Rollout: A controlled release of a feature to a selected percentage or cohort of users.

Environment override: A flag value or setting configured independently for an environment such as Development, Testing, or Production.

Client SDK: A software library integrated into an application to request flag evaluations and receive configuration updates.

2. Overall Description

2.1 Product Perspective

The platform acts as a central configuration service used by management users and client applications. Management users update flags and rollout settings through the platform; client SDKs request evaluations and receive updated configuration states.

2.2 User Classes

- Software Engineer: creates and manages flags, defines targeting rules, configures environment overrides, and integrates the client SDK.
- Release Manager: configures and monitors percentage-based rollouts and initiates rollbacks when needed.
- Client SDK (external system actor): requests flag evaluations and receives configuration updates on behalf of an application.

2.3 Operating Environment

The project is specified as a software platform with a management interface, a flag-evaluation service, configuration storage, and client SDK integration. Specific programming languages, frameworks, hosting infrastructure, and database products are not prescribed in this SRS and may be selected during design.

2.4 Constraints and Assumptions

- Each feature flag has a unique identifier/key.
- Targeting decisions depend on the configured rules and the evaluation request context.
- Development, Testing, and Production configurations must be independently manageable.

- The evaluation API target is under 15 ms, as specified in NFR-001; the measurement conditions should be defined during testing.
- Real-time synchronization requires connected clients and an agreed synchronization mechanism/interval, to be finalized during design.

3. Specific Requirements

3.1 Functional Requirements

ID	Requirement
FR-001	The system shall evaluate user targeting rules (e.g., User ID hash, Beta cohort) and return accurate boolean feature flag states.
FR-002	The system shall allow authorized users to create, view, update, and deactivate feature flags, each with a unique identifier.
FR-003	The system shall allow Release Managers to configure percentage-based rollouts for selected user cohorts.
FR-004	The system shall support environment-specific configuration overrides for Development, Testing, and Production environments.
FR-005	The system shall synchronize updated feature flag states with connected client SDKs in real time.

3.2 Non-Functional Requirements

ID	Type	Requirement
NFR-001	Performance & Security	The feature flag evaluation API must execute in under 15 ms to minimize overhead when integrated into client critical paths. Security expectations and benchmark conditions must be defined and verified during testing.
NFR-002	Security	The system shall restrict feature-flag and configuration-management operations to authenticated users with appropriate permissions.

4. External Interface Requirements

4.1 User Interface

The management interface should provide views and actions for creating, viewing, updating, and deactivating flags; configuring targeting and environment overrides; and setting and monitoring rollouts. Exact screen layouts are outside the current scope.

4.2 Software Interfaces

The system shall expose a client-facing evaluation interface for SDK requests and a management interface for authorized configuration operations. The protocol, endpoint format, authentication mechanism, and SDK update transport are design decisions to be specified before implementation.

5. Use Cases

- UC-01 Manage Feature Flags
- UC-02 Configure Targeting Rules
- UC-03 Evaluate Feature Flag
- UC-04 Configure Percentage Rollout
- UC-05 Manage Environment Overrides
- UC-06 Synchronize Client SDK
- UC-07 Monitor Rollout
- Supporting use cases: Validate Targeting Rules; Roll Back Rollout.

6. Verification and Traceability

Each requirement is associated with a verification activity in the Requirements Traceability Matrix (RTM). The RTM is maintained in the companion Requirements and RTM workbook. Test cases are planned activities and must not be marked passed until executed and supported by results.

7. Items to Be Finalized

- Define the exact benchmark load, hardware/runtime environment, and latency measurement method for NFR-001.
- Specify the authentication and role-permission model for management actions.
- Define the precise behavior when a flag is missing, evaluation context is incomplete, or the service is unavailable.
- Define the synchronization mechanism and acceptable propagation interval for client SDK updates.